

Assignment - 02

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Section: 3

Given grammar,

1. $D \rightarrow T \text{ id } ; D$

2. $D \rightarrow \epsilon$

3. $T \rightarrow B \epsilon$

4. $B \rightarrow \text{int}$

5. $B \rightarrow \text{slot}$

6. $C \rightarrow [\text{num}] \epsilon$

7. $C \rightarrow \epsilon$

SLR(1) parse table -

given input string - $\text{int}[5]a;$

Lexeme	int	[	5	]	a	;
Token	int	[	num	]	id	,

So, the token stream is: $\text{int}[\text{num}] \text{id} ; \$$

Step	Stack	Symbol	Input	Action
1	0	\$	int[num]id ; \$	Shift to 4
2	04	\$int	[num]id ; \$	Reduce by rule 4: $B \rightarrow \text{int}$
3	03	\$B	[num]id ; \$	Shift to 8
4	038	\$B[	num]id ; \$	Shift to 10
5	03810	\$B[num	]id ; \$	Shift to 12
6	0381012	\$B[num]	id ; \$	Reduce by rule 7: $C \rightarrow \epsilon$
7	038101213	\$B[num]C	id ; \$	Reduce by rule 6: $C \rightarrow [\text{num}]C$
8	037	\$BC	id ; \$	Reduce by rule 3: $T \rightarrow BC$
9	02	\$T	id ; \$	Shift to 6
10	026	\$Tid	; \$	Shift to 9
11	0269	\$Tid ;	\$	Reduce by rule 2: $D \rightarrow \epsilon$
12	026911	\$Tid ; D	\$	Reduce by rule 1: $D \rightarrow T \text{ id } ; D$
13	01	\$D	\$	Accept

So it can be said that $\text{int}[5]a;$ is a valid input string according to the given grammar.